

Innovative Mathematics Olympiad
Model Paper 2025 – 2026

Total Questions: 50

Time: 1.5 hr.

Total Marks: 100

Syllabus

Real Numbers, Polynomials, Pair of Linear Equations, Quadratic Equations, Progressions, Triangles, Coordinate Geometry, Introduction to Trigonometry, Applications of Trigonometry, Circles, Surface Areas and Volumes, Statistics, Probability.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Mathematics

1. If a positive integer when divided by 7 leaves a remainder 3, which of the following must always be divisible by 7?
 (A) $n^2 + 4n + 5$ (B) $n^2 - 2n - 3$
 (C) $n^2 + n - 10$ (D) $n^2 - 8$
2. A polynomial $p(x)$ leaves remainders 3 and 5 when divided by $(x-1)$ and $(x-2)$ respectively. What is the remainder when $p(x)$ is divided by $(x-1)(x-2)$?
 (A) $2x + 1$ (B) $x + 2$
 (C) $2x + 3$ (D) $x + 3$
3. **Statement I:** The square of any even integer is divisible by 4.
Statement II: Every even integer can be expressed in the form $2n$, where n is an integer.
 (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
- (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true
4. Two numbers have a sum of 27. Three times the first number exceeds twice the second by 4. What is the product of the two numbers?
 (A) 120 (B) 166.4
 (C) 130 (D) 178.64
5. If the roots of the quadratic equation $x^2 - (k+3)x + (k+2) = 0$ are equal, then the value of k is:
 (A) -1 (B) 1
 (C) 2 (D) 3
6. The sum of first n terms of an arithmetic progression is $3n^2 + 5n$. What is the 10th term of the A.P.?
 (A) 62 (B) 65
 (C) 68 (D) 70

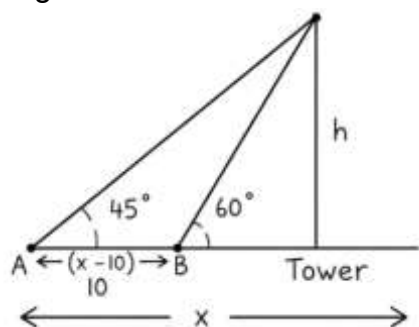
7. In two similar triangles, the ratio of their areas is 9: 25. What is the ratio of their corresponding sides?

(A) 3: 5 (B) 9: 25
(C) 5: 3 (D) 25: 9

8. The coordinates of points A and B are (2, -2) and (8, 4). A point P divides AB internally in the ratio 1: 2. What is the distance of P from the origin?

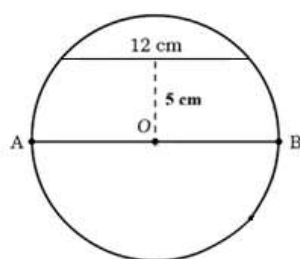
(A) 2 (B) 4
(C) $\sqrt{20}$ (D) $\sqrt{26}$

9. The angle of elevation of the top of a tower from a point on the ground is 45° . If the point is moved 10 m closer to the tower, the angle becomes 60° . What is the height of the tower?



(A) $12 + 5\sqrt{3}$ (B) $15 + 5\sqrt{3}$
(C) $4 + 10\sqrt{3}$ m (D) $5 + 10\sqrt{3}$ m

10. A chord of a circle is 12 cm long and is at a distance 5 cm from the centre. What is the radius of the circle?



(A) $\sqrt{67}$ cm (B) $\sqrt{65}$ cm
(C) $\sqrt{61}$ cm (D) $\sqrt{64}$ cm

11. The curved surface area of a cylinder is equal to twice its base.

If the radius of the base is 7 cm, find the height of the cylinder.

(A) 7 cm (B) 14 cm
(C) 21 cm (D) 28 cm

12. The HCF of two positive integers is 12 and their product is 1728. What is the number of pairs of such integers possible?

(A) 1 (B) 2
(C) 3 (D) 4

13. If α and β are the zeroes of the polynomial $x^2 - 7x + 10$ then the value of $\alpha^2 + \beta^2$ is:

(A) 29 (B) 39
(C) 49 (D) 59

14. A shopkeeper sells two articles for ₹240 each. On one he gains 20% and on the other he loses 20%. What is his overall gain or loss?

(A) Gain of ₹20
(B) Loss of ₹20
(C) No gain, no loss
(D) Gain of ₹40

15. For what value of k does the quadratic equation $kx^2 - 4x + 1 = 0$ have exactly one real root?

(A) 1 (B) 2
(C) 4 (D) 8

16. If the 7th term of an A.P. is 18 and 13th term is 30, what is the sum of the first 20 terms of the A.P?

(A) 380 (B) 500
(C) 420 (D) 440

17. In a right-angled triangle, if the altitude drawn to the hypotenuse divides it into segments of lengths 9 cm and 16 cm, then the length of the hypotenuse is:

- (A) 25 cm (B) 24 cm
(C) 20 cm (D) 13 cm

18. The coordinates of vertices of a triangle are (1, 2), (4, 6), and (7, 10). The area of the triangle is:

- (A) 0 (B) 6
(C) 12 (D) 24

19. The information is given in which alternative is incorrectly matched?

Statistical Term	Property
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- | | |
|------------|---|
| (A) Mean | Sum of all observations divided by the number of observations |
| (B) Median | Middle observation when data is arranged in order |
| (C) Mode | Observation that occurs least frequently |
| (D) Range | Difference between the greatest and the least values |

20. If $\sin \theta = \frac{5}{13}$, θ acute then the value of $\frac{1 - \cos \theta}{1 + \cos \theta}$ is:

- (A) $\frac{1}{25}$ (B) $\frac{4}{9}$
(C) $\frac{9}{4}$ (D) $\frac{1}{34}$

21. The mean of 10 observations is 25. If one observation is removed, the mean becomes 24. What is the removed observation?

- (A) 15 (B) 20
(C) 25 (D) 34

22. Two unbiased dice are thrown. What is the probability that the sum is a prime number?

- (A) $\frac{5}{12}$ (B) $\frac{1}{3}$
(C) $\frac{7}{36}$ (D) $\frac{5}{18}$

23. If n is a positive integer such that n^2 is divisible by 3, which of the following must be true?

- (A) n is divisible by 3
(B) n is even
(C) $n + 1$ is divisible by 3
(D) n^2 is divisible by 8

24. If $p(x)$ is a cubic polynomial such that the sum of its zeroes is 0 and the product of its zeroes is 6, then which of the following can be $p(x)$?

- (A) $x^3 - 6$ (B) $x^3 + 6$
(C) $x^3 - 3x - 5$ (D) $x^3 + 3x + 6$

25. Two numbers differ by 16. The sum of twice the smaller number and thrice the larger number is 182. What is the larger number?

- (A) 36 (B) 42.8
(C) 52.5 (D) 60

26. For what value of k does the quadratic equation $x^2 + 2kx + (2k^2 - 9) = 0$ have real and equal roots?

- (A) 2 (B) -2
(C) 3 (D) -4

27. If the arithmetic mean between two positive numbers is 20 and their geometric mean is 16, what is the difference between the numbers?

- (A) 8 (B) 12
(C) 16 (D) 24

28. In a triangle, a line parallel to the base divides the other two sides in the ratio 3:5. What is the ratio of the area of the smaller triangle to the larger triangle?
 (A) 3:5 (B) 9:64
 (C) 5:8 (D) 8:25
29. Find the coordinates of the point which divides the line joining (6, -2) and (-4, 8) externally in the ratio 1:2.
 (A) (2, 4)
 (B) (10, -6)
 (C) (8, -8)
 (D) (16, -12)
30. If $\tan \theta + \sec \theta = 4$ where θ is acute, then the value of $\tan \theta - \sec \theta$ is:
 (A) -1/4 (B) -2
 (C) 3/4 (D) 2
31. The mean of 5 observations is 18. If three more observations 10, 14, and 20 are added, what is the new mean?
 (A) 15 (B) 16.75
 (C) 17 (D) 18.5
32. A card is drawn at random from a well-shuffled pack of 52 cards. What is the probability that the card drawn is neither a king nor a heart?
 (A) $\frac{9}{13}$ (B) $\frac{10}{13}$
 (C) $\frac{11}{13}$ (D) $\frac{12}{13}$
33. If $x = 2$ is a zero of the polynomials $p(x) = ax^3 - 4x^2 + bx - 8$ and the sum of its zeroes is 4, then the value of $a + b$ is:
 (A) 2 (B) 4
 (C) 6 (D) 9
34. If a positive integer n leaves remainder 1 when divided by 4, which of the following is always divisible by 8?
 (A) $n^2 - 1$ (B) $n^2 + 1$
 (C) $n^2 - 5$ (D) $n^2 + 3$
35. The sum of the digits of a two-digit number is 12. The number formed by reversing the digits exceeds the original number by 36. What is the original number?
 (A) 48 (B) 39
 (C) 57 (D) 66
36. For what value of k does the quadratic equation $(k - 1)x^2 + 6x + (k + 2) = 0$ have no real roots?
 (A) 0 (B) 1
 (C) 2 (D) 3
37. The sum of the first n terms of an arithmetic progression is $S_n = 2n^2 + 3n$. What is the common difference of the A.P.?
 (A) 2 (B) 3
 (C) 4 (D) 5
38. In a triangle, the ratio of the sides is 3:4:5. The ratio of the areas of the squares constructed on the largest and smallest sides is:
 (A) 25:9 (B) 9:25
 (C) 5:3 (D) 3:5
39. The centroid of a triangle is at (2, 3). Two vertices of the triangle are (1, 2) and (5, 4). What are the coordinates of the third vertex?
 (A) (0, 3) (B) (0, 5)
 (C) (2, 4) (D) (4, 3)

40. If $\sin \theta = \cos \theta$ and $0^\circ < \theta < 90^\circ$, then the value of $\sin 2\theta$ is:
 (A) 0 (B) $\frac{1}{2}$
 (C) 1 (D) $\frac{\sqrt{3}}{2}$
41. The mean of 15 observations is 20. If the mean of the first 8 observations is 18 and that of the last 8 observations is 22, find the 8th observation.
 (A) 18 (B) 19
 (C) 20 (D) 22
42. Two dice are thrown simultaneously. What is the probability that the difference of the numbers on the top is 2?
 (A) $\frac{1}{6}$ (B) $\frac{2}{9}$
 (C) $\frac{5}{18}$ (D) $\frac{1}{9}$
43. Two concentric circles have radii in the ratio 2:3. What is the ratio of the areas of the regions between the smaller and the larger circles?
 (A) 4:9 (B) 5:9
 (C) 1:5 (D) 9:4
44. A chord of a circle is 10 cm long and is 12 cm away from the centre. The diameter of the circle is:
 (A) 20 cm (B) 24 cm
 (C) 26 cm (D) 28 cm
45. A solid sphere and a solid cube are made of the same material. If their volumes are equal and the edge of the cube is 6 cm, the radius of the sphere is:
 (A) $\frac{3}{2} \sqrt[3]{\pi}$ (B) $\sqrt[3]{\frac{162}{\pi}}$
 (C) $\sqrt[3]{\frac{216}{\pi}}$ (D) $\sqrt[3]{\frac{108}{\pi}}$
46. The curved surface area of a cone is 308 cm^2 and its slant height is 14 cm. What is the radius of the base?
 (A) 7 cm (B) 8 cm
 (C) 10 cm (D) 11 cm
47. The mean of 12 observations is 18. If three observations 10, 15 and 20 are removed, what is the new mean?
 (A) 17 (B) 18
 (C) 19 (D) 20
48. In a frequency distribution, the class marks are 15, 20, 25, The class width is:
 (A) 2.5 (B) 5
 (C) 10 (D) 15
49. A bag contains 5 red and 7 blue balls. Two balls are drawn without replacement. What is the probability that both are blue?
 (A) $\frac{7}{12}$ (B) $\frac{21}{66}$
 (C) $\frac{49}{144}$ (D) $\frac{7}{11}$
50. If two events A and B are such that $P(A) = \frac{3}{5}$, $P(B) = \frac{2}{5}$, $A \cap B = \emptyset$ then $P(A \cup B)$ is:
 (A) $\frac{1}{5}$ (B) $\frac{2}{5}$
 (C) $\frac{3}{5}$ (D) 1